```c
// A program that says hello to the world

#include <stdio.h>

int main(void)
{
    printf("hello, world\n");
}
```

```c
// get_string and printf with incorrect placeholder

#include <cs50.h>
#include <stdio.h>

int main(void)
{
    string answer = get_string("What's your name? ");
    printf("hello, answer\n");
}
```

```c
// get_string and printf with %s

#include <cs50.h>
#include <stdio.h>

int main(void)
{
    string answer = get_string("What's your name? ");
    printf("hello, %s\n", answer);
}
```

```c
// Conditional, Boolean expression, relational operator

#include <cs50.h>
#include <stdio.h>

int main(void)
{
    // Prompt user for integers
    int x = get_int("What's x? ");
    int y = get_int("What's y? ");

    // Compare integers
    if (x < y)
    {
        printf("x is less than y\n");
    }
}
```

```c
// Conditionals that are mutually exclusive

#include <cs50.h>
#include <stdio.h>

int main(void)
{
    // Prompt user for integers
    int x = get_int("What's x? ");
    int y = get_int("What's y? ");

    // Compare integers
    if (x < y)
    {
        printf("x is less than y\n");
    }
    else
    {
        printf("x is not less than y\n");
    }
}
```

```c
// Conditionals that aren't mutually exclusive

#include <cs50.h>
#include <stdio.h>

int main(void)
{
    // Prompt user for integers
    int x = get_int("What's x? ");
    int y = get_int("What's y? ");

    // Compare integers
    if (x < y)
    {
        printf("x is less than y\n");
    }
    if (x > y)
    {
        printf("x is greater than y\n");
    }
    if (x == y)
    {
        printf("x is equal to y\n");
    }
}
```

```c
// Conditional that isn't necessary

#include <cs50.h>
#include <stdio.h>

int main(void)
{
    // Prompt user for integers
    int x = get_int("What's x? ");
    int y = get_int("What's y? ");

    // Compare integers
    if (x < y)
    {
        printf("x is less than y\n");
    }
    else if (x > y)
    {
        printf("x is greater than y\n");
    }
    else if (x == y)
    {
        printf("x is equal to y\n");
    }
}
```

```c
// Conditionals

#include <cs50.h>
#include <stdio.h>

int main(void)
{
    // Prompt user for integers
    int x = get_int("What's x? ");
    int y = get_int("What's y? ");

    // Compare integers
    if (x < y)
    {
        printf("x is less than y\n");
    }
    else if (x > y)
    {
        printf("x is greater than y\n");
    }
    else
    {
        printf("x is equal to y\n");
    }
}
```

```c
// Comparing against lowercase char

#include <cs50.h>
#include <stdio.h>

int main(void)
{
    // Prompt user to agree
    char c = get_char("Do you agree? ");

    // Check whether agreed
    if (c == 'y')
    {
        printf("Agreed.\n");
    }
    else if (c == 'n')
    {
        printf("Not agreed.\n");
    }
}
```

```c
// Comparing against lowercase and uppercase char

#include <cs50.h>
#include <stdio.h>

int main(void)
{
    // Prompt user to agree
    char c = get_char("Do you agree? ");

    // Check whether agreed
    if (c == 'y')
    {
        printf("Agreed.\n");
    }
    else if (c == 'Y')
    {
        printf("Agreed.\n");
    }
    else
    {
        printf("Not agreed.\n");
    }
}
```

```c
// Logical operators

#include <cs50.h>
#include <stdio.h>

int main(void)
{
    // Prompt user to agree
    char c = get_char("Do you agree? ");

    // Check whether agreed
    if (c == 'Y' || c == 'y')
    {
        printf("Agreed.\n");
    }
    else
    {
        printf("Not agreed.\n");
    }
}
```

```c
// Opportunity for better design

#include <stdio.h>

int main(void)
{
    printf("meow\n");
    printf("meow\n");
    printf("meow\n");
}
```

```c
// Better design

#include <stdio.h>

int main(void)
{
    int i = 3;
    while (i > 0)
    {
        printf("meow\n");
        i--;
    }
}
```

```c
// Print values of i

#include <stdio.h>

int main(void)
{
    int i = 1;
    while (i <= 3)
    {
        printf("meow\n");
        i++;
    }
}
```

```c
// Better design

#include <stdio.h>

int main(void)
{
    int i = 0;
    while (i < 3)
    {
        printf("meow\n");
        i++;
    }
}
```

```c
// Better design

#include <stdio.h>

int main(void)
{
    for (int i = 0; i < 3; i++)
    {
        printf("meow\n");
    }
}
```

```c
// Infinite loop

#include <cs50.h>
#include <stdio.h>

int main(void)
{
    while (true)
    {
        printf("meow\n");
    }
}
```

```c
// Prompts user for n.

#include <cs50.h>
#include <stdio.h>

int main(void)
{
    int n = get_int("What's n? ");

    for (int i = 0; i < n; i++)
    {
        printf("meow\n");
    }
}
```

```c
// Prompts user again if need be. (Poor design.)

#include <cs50.h>
#include <stdio.h>

int main(void)
{
    int n = get_int("What's n? ");
    if (n < 0)
    {
        n = get_int("What's n? ");
    }

    for (int i = 0; i < n; i++)
    {
        printf("meow\n");
    }
}
```

```c
// Uses a loop with continue/break.

#include <cs50.h>
#include <stdio.h>

int main(void)
{
    int n;
    while (true)
    {
        n = get_int("What's n? ");
        if (n < 0)
        {
            continue;
        }
        else
        {
            break;
        }
    }

    for (int i = 0; i < n; i++)
    {
        printf("meow\n");
    }
}
```

```c
// Uses a loop with just break.

#include <cs50.h>
#include <stdio.h>

int main(void)
{
    int n;
    while (true)
    {
        n = get_int("What's n? ");
        if (n >= 0)
        {
            break;
        }
    }

    for (int i = 0; i < n; i++)
    {
        printf("meow\n");
    }
}
```

```c
// Uses a do-while loop instead.

#include <cs50.h>
#include <stdio.h>

int main(void)
{
    int n;
    do
    {
        n = get_int("What's n? ");
    }
    while (n < 0);

    for (int i = 0; i < n; i++)
    {
        printf("meow\n");
    }
}
```

```c
// Abstraction

#include <stdio.h>

void meow(void);

int main(void)
{
    for (int i = 0; i < 3; i++)
    {
        meow();
    }
}

// Meow once
void meow(void)
{
    printf("meow\n");
}
```

```c
// Abstraction with parameterization

#include <stdio.h>

void meow(int n);

int main(void)
{
    meow(3);
}

// Meow some number of times
void meow(int n)
{
    for (int i = 0; i < n; i++)
    {
        printf("meow\n");
    }
}
```

```c
// Demonstrates scope

#include <stdio.h>

void meow(int n);

int main(void)
{
    int n = 3;
    meow(n);
}

// Meow some number of times
void meow(int n)
{
    for (int i = 0; i < n; i++)
    {
        printf("meow\n");
    }
}
```

```c
// User input

#include <cs50.h>
#include <stdio.h>

void meow(int n);

int main(void)
{
    int n;
    do
    {
        n = get_int("Number: ");
    }
    while (n < 1);
    meow(n);
}

// Meow some number of times
void meow(int n)
{
    for (int i = 0; i < n; i++)
    {
        printf("meow\n");
    }
}
```

```c
// Return value

#include <cs50.h>
#include <stdio.h>

int get_positive_int(void);
void meow(int n);

int main(void)
{
    int n = get_positive_int();
    meow(n);
}

// Get number of meows
int get_positive_int(void)
{
    int n;
    do
    {
        n = get_int("Number: ");
    }
    while (n < 1);
    return n;
}

// Meow some number of times
void meow(int n)
{
    for (int i = 0; i < n; i++)
    {
        printf("meow\n");
    }
}
```

```c
// Prints a row of 4 question marks

#include <stdio.h>

int main(void)
{
    printf("????\n");
}
```

```c
// Prints a row of 4 question marks with a loop

#include <stdio.h>

int main(void)
{
    for (int i = 0; i < 4; i++)
    {
        printf("?");
    }
    printf("\n");
}
```

```c
// Prints a column of 3 bricks with a loop

#include <stdio.h>

int main(void)
{
    for (int i = 0; i < 3; i++)
    {
        printf("#\n");
    }
}
```

```c
// Prints a 3-by-3 grid of bricks with nested loops

#include <stdio.h>

int main(void)
{
    for (int i = 0; i < 3; i++)
    {
        for (int j = 0; j < 3; j++)
        {
            printf("#");
        }
        printf("\n");
    }
}
```

```c
// Prints a 3-by-3 grid of bricks with nested loops using a constant

#include <stdio.h>

int main(void)
{
    const int n = 3;
    for (int i = 0; i < n; i++)
    {
        for (int j = 0; j < n; j++)
        {
            printf("#");
        }
        printf("\n");
    }
}
```

```c
// Helper function

#include <stdio.h>

void print_row(int width);

int main(void)
{
    const int n = 3;
    for (int i = 0; i < n; i++)
    {
        print_row(n);
    }
}

void print_row(int width)
{
    for (int i = 0; i < width; i++)
    {
        printf("#");
    }
    printf("\n");
}
```

```c
// Addition with int

#include <cs50.h>
#include <stdio.h>

int main(void)
{
    // Prompt user for x
    int x = get_int("What's x? ");

    // Prompt user for y
    int y = get_int("What's y? ");

    // Add numbers
    int z = x + y;

    // Perform addition
    printf("%i\n", z);
}
```

```c
// Addition with int, without third variable

#include <cs50.h>
#include <stdio.h>

int main(void)
{
    // Prompt user for x
    int x = get_int("What's x? ");

    // Prompt user for y
    int y = get_int("What's y? ");

    // Perform addition
    printf("%i\n", x + y);
}
```

```c
// Doubles a number

#include <cs50.h>
#include <stdio.h>

int main(void)
{
    // Prompt user for x
    int x = get_int("What's x? ");

    // Double it
    printf("%i\n", x * 2);
}
```

```c
// Overflow

#include <cs50.h>
#include <stdio.h>

int main(void)
{
    int dollars = 1;
    while (true)
    {
        char c = get_char("Here's $%i. Double it and give to next person? ", dollars);
        if (c == 'y')
        {
            dollars *= 2;
        }
        else
        {
            break;
        }
    }
    printf("Here's $%i.\n", dollars);
}
```

```c
// long

#include <cs50.h>
#include <stdio.h>

int main(void)
{
    long dollars = 1;
    while (true)
    {
        char c = get_char("Here's $%li. Double it and give to next person? ", dollars);
        if (c == 'y')
        {
            dollars *= 2;
        }
        else
        {
            break;
        }
    }
    printf("Here's $%li.\n", dollars);
}
```

```c
// Division with ints, demonstrating truncation

#include <cs50.h>
#include <stdio.h>

int main(void)
{
    // Prompt user for x
    int x = get_int("What's x? ");

    // Prompt user for y
    int y = get_int("What's y? ");

    // Divide x by y
    printf("%i\n", x / y);
}
```

```c
// Casting

#include <cs50.h>
#include <stdio.h>

int main(void)
{
    // Prompt user for x
    int x = get_int("What's x? ");

    // Prompt user for y
    int y = get_int("What's y? ");

    // Divide x by y
    printf("%f\n", (float) x / y);
}
```

```
// Floats

#include <cs50.h>
#include <stdio.h>

int main(void)
{
    // Prompt user for x
    float x = get_float("What's x? ");

    // Prompt user for y
    float y = get_float("What's y? ");

    // Divide x by y
    printf("%.50f\n", x / y);
}
```